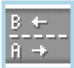
Image



Description

You might, for example, use the *TwoLaneTrack* and the *Transporter*to model an AGV (automated guided vehicle) system. The distance, which the **Transporter** has to travel on the *TwoLaneTrack* is defined by the **Length** of **Lane A** and the **Length** of **Lane B** of the *TwoLaneTrack*, the *Transporter's* **MU Length**, and its **Speed**. They determine the time, which the *Transporter* remains on the *TwoLaneTrack*. As opposed to the point-oriented material flow objects, *Plant Simulation* uses the actual length, which you type in, during your simulation run. A *Transporter* may not pass another one moving in front of it. The *Transporters* thus retain their order of moving onto and of leaving the *TwoLaneTrack*. Each lane of the *TwoLaneTrack* may have its own length to realistically model the length of the lanes, when the *TwoLaneTrack* turns the corner. In that case the outside lane is longer than the inside lane.

If several *Transporters* travel along a lane of the *TwoLaneTrack* at a different speed, the faster one collides with the slower one. *Plant Simulation* then activates the **Collision Control** of the faster *Transporter* and automatically reduces its speed to the speed of the slower *Transporter*.

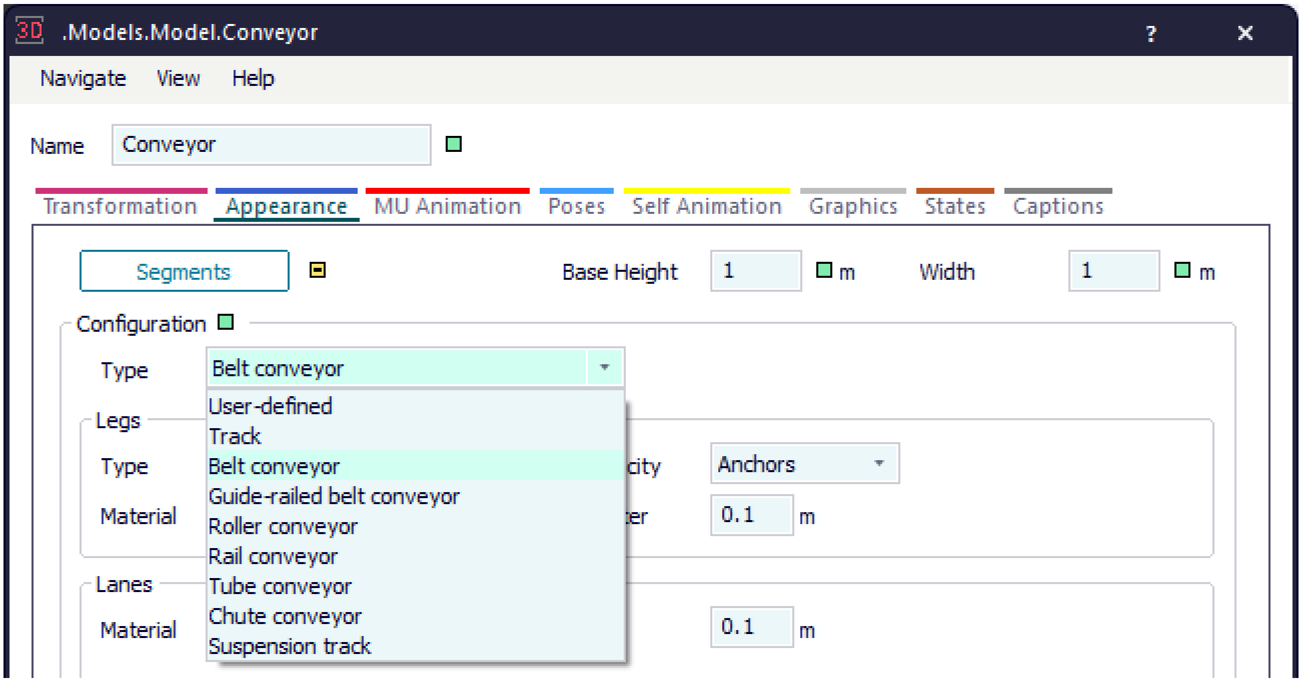
The *Transporter* can drive forward and **Backwards** on the *TwoLaneTrack*, i.e., it drives onto the *TwoLaneTrack* at its **exit** and exits it at its **entrance**. Driving forward and backward are not properties of the *TwoLaneTrack*, but of the *Transporters*, which drive on it.

The maximum capacity of the *TwoLaneTrack* is defined by its length and the lengths of the individual *Transporters* moving on it, i.e., a *TwoLaneTrack* that is three yards long accepts three *Transporters* of one yard each at the most. With the **Capacity** you can further restrict the number of *Transporters* located on the *TwoLaneTrack*.

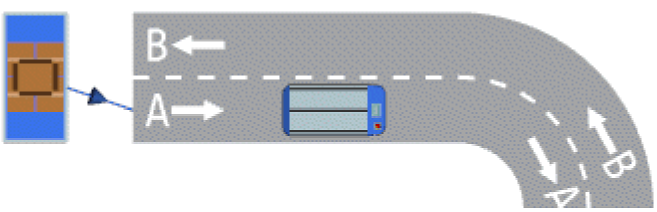
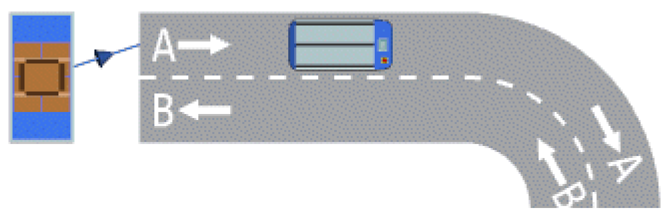
You can insert the *TwoLaneTrack*:

- As a curved object, which is the default setting.
- By inserting any sequence of curved segments and straight segments, you can realistically model curved conveyor systems on which the *Transporters* move.

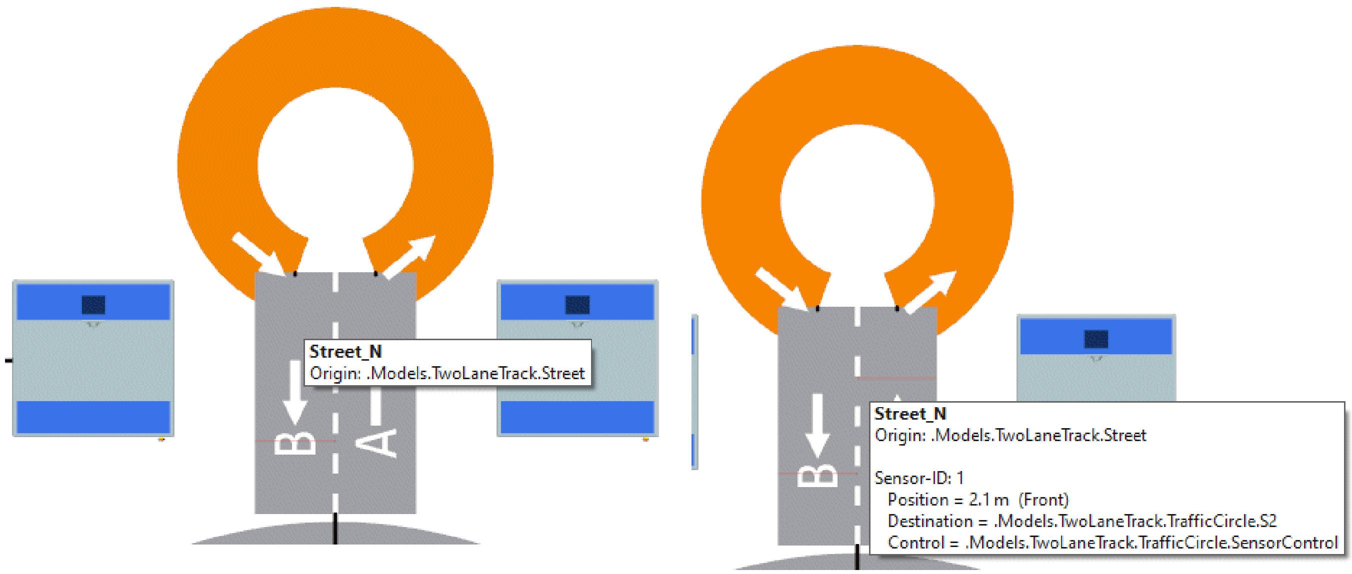
  You can select different configurations for the *TwoLaneTrack* on the tab **Appearance**.



The *TwoLaneTrack* supports bidirectional traffic. You can thus select if **Right-hand Traffic** or **Left-hand Traffic** applies.

Right-hand traffic	Left-hand traffic
	

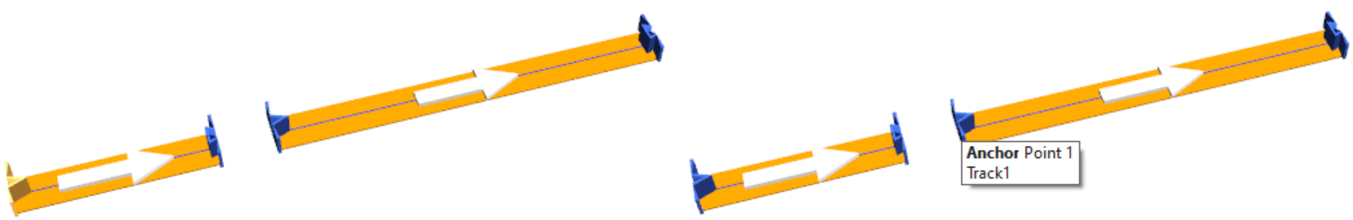
To show a **tooltip** with information about the *TwoLaneTrack*, hover with the mouse over it.



Show Manipulators

To change the length of the graphic and the anchor points of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

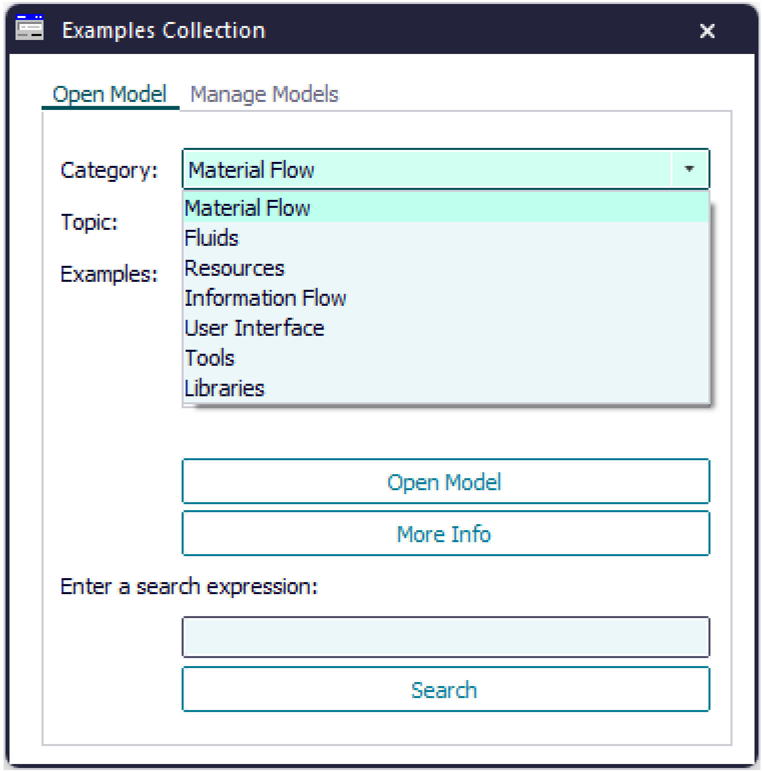
The manipulator at the start and at the end of the length-oriented object is cut off. When you attach an object of the same type to the other object, the two combined halves result in a complete manipulator again. When you move the mouse over a manipulator, *Plant Simulation* shows a *Tooltip*.



Add the Object to the Simulation Model

To add the object *TwoLaneTrack* to your simulation model, click **Library** > **Basic Objects > MaterialFlow > TwoLaneTrack** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

Model a Transport System with Passive Objects

Work with Length-oriented Objects

Routing [TwoLaneTrack]

Routing [TwoLaneTrack]

To facilitate a branching *TwoLaneTrack*, use any of these controls to move a *Transporter* on to the succeeding object.

Remarks

Plant Simulation uses:

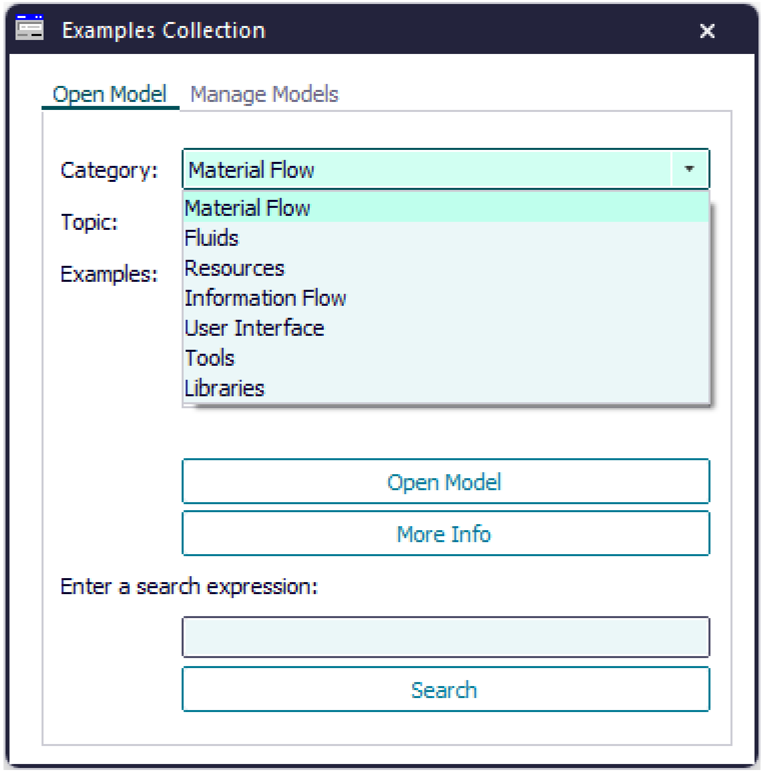
- The **Exit Control** of the *TwoLaneTrack*.
- **Automatic Routing** of the *Transporter*.
- The **Driving Control** of the *Transporter*.
- The built-in properties of the *TwoLaneTrack* described below.

The **Exit Control** has the highest priority. If you did not define an **Exit Control** for the *TwoLaneTrack* and if you typed in a destination list for the *Transporter*, **Automatic Routing** moves the *Transporter* on to the proper successor, using the destination list of the successors.

You can also assign a **Destination** to the *Transporter*. The *TwoLaneTrack* has a *Forward destination list* and *Backward destination list* for each of its lanes, into which you type all destinations that the *Transporter* can reach, when it moves on the respective lane of the *TwoLaneTrack*. If you assigned a destination list to a lane of the *Transporter* that is ready to be moved on, the *TwoLaneTrack* searches the *Forward destination list* and the *Backward destination list* of all **successors** for the *Destination* you typed in, and, as soon as it finds it, moves the *Transporter* on to that successor. If the *TwoLaneTrack* does not find the destination, it moves the *Transporter* to the next successor in turn. *Plant Simulation* terminates the search after it finds the first instance of the *Destination*. Automatic routing is inactive if you did not type in a *Destination* for the *Transporter* or if you typed in an **Exit Control** for the *TwoLaneTrack*.

If you did not type in an **Exit Control** nor any **Destination Lists** for the successors, the *TwoLaneTrack* uses the **Driving Control** of the *Transporter* to move it. If you do not type in any controls at all, the *TwoLaneTrack* moves the *Transporters* to each of the successors, to which it is connected, in turn.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

Model a Transport System with Passive Objects

Dialog Box of the TwoLaneTrack

Dialog Box of the TwoLaneTrack

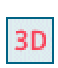
Double-click the icon of the *TwoLaneTrack* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box **Edit 3D Properties**:

- Click the button 
 Edit 3D Properties in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

See also

Tab Controls A/B

Entrance Locked [lane A]

Exit Locked [lane A]

Entrance Locked [lane B]

Exit locked [lane B]

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

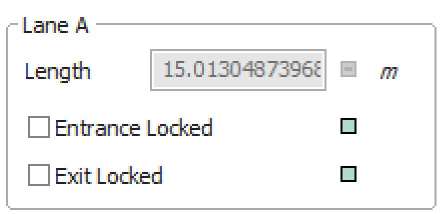
The shared properties are described under the **Tab Attributes**.

Lane A

Lane A

Lane A provides a number of settings.

Length, **Entrance Locked**, and **Exit Locked**.



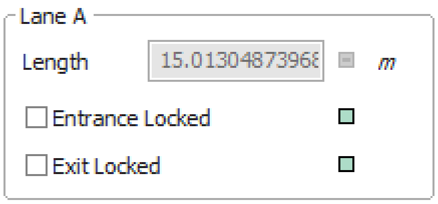
See also

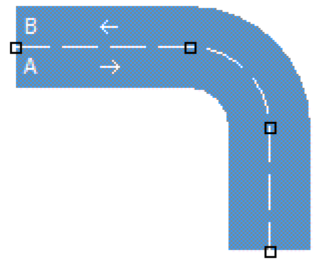
asAny [SimTalk]

IsLaneA [SimTalk]

Length [text box] - lane A

Type the **Length** of **Lane A** of the *TwoLaneTrack* into the text box.





Remarks

After you inserted it, *Plant Simulation* shows its length here. The length of the lanes of the *TwoLaneTrack* and the sum of the lengths of all *Transporters* located on it determine how many *Transporters* it can accommodate at any one time.

Each lane of the *TwoLaneTrack* may have its own length to realistically model the length of the lanes, when the *TwoLaneTrack* turns a corner. In that case the outside lane is longer than the inside lane.

You can change the length of the graphic and the anchor points of the *TwoLaneTrack* by clicking **Show Manipulators**  on the **Edit** ribbon tab or by pressing **M** on the keyboard.

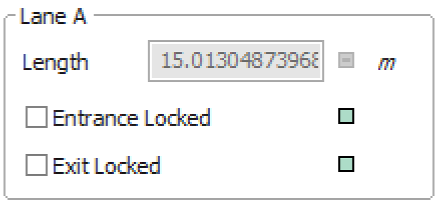
SimTalk

A.**Length [SimTalk] - lane A or B**

A.**OccupiedLength [SimTalk] - lane A or B**

Entrance Locked [lane A]

To prevent that *Transporters* enter **lane A** of the *TwoLaneTrack*, select this check box.



Remarks

The *TwoLaneTrack* will finish transporting the *Transporters* that are already located on it.

It does not accept any additional *Transporters* and enters those into the **Blocking List** of **Lane A**.

- If the *Transporter* wants to enter the *TwoLaneTrack* at its entrance, it is entered into the **Forward Blocking List** of **Lane A**.
- If the *Transporter* is moving backward and wants to enter a length-oriented object at its exit, it is entered into the **Exit Blocking List** of **Lane A**.

Transporting starts again once you clear the check box. The first *Transporter* in the **Forward Blocking List** of **Lane A** is going to be moved on first.

SimTalk

A.**EntranceLocked [SimTalk] - lane A or B**

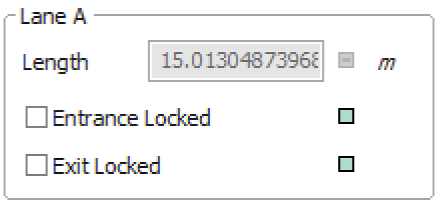
See also

Forward Blocking List

Exit Blocking List

Exit Locked [lane A]

To prevent that *Transporters* exit **Lane A** of the *TwoLaneTrack*, select this check box.



Remarks

The *TwoLaneTrack* does not transport the *Transporters* to the successor in the material flow, but enters them into the **Exit Blocking List** of **Lane A** of the *TwoLaneTrack*.

Once the exit is unlocked again, the *TwoLaneTrack* moves the first *Transporter* in the **Exit Blocking List** of **Lane A** on to its successor.

Note

The *Transporter* does not trigger the **Exit Control** if you selected the check box **Exit Locked**.

SimTalk

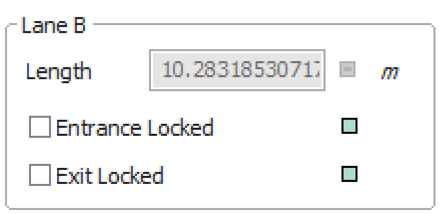
A.**ExitLocked [SimTalk] - lane A or B**

Lane B

Lane B

Lane A provides a number of settings.

Length, **Entrance Locked**, and **Exit Locked**.



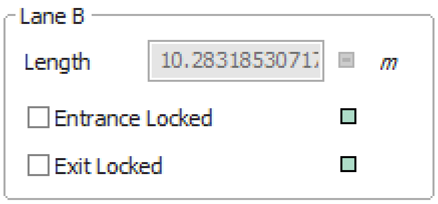
See also

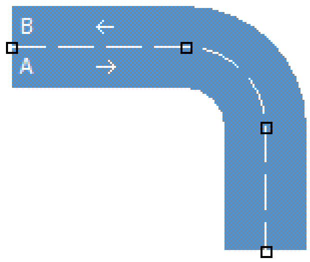
asAny [SimTalk]

IsLaneB [SimTalk]

Length [text box] - lane B

Type the **Length** of **Lane B** of the *TwoLaneTrack* into the text box.





Remarks

After you inserted it, *Plant Simulation* shows its length here. The length of the lanes of the *TwoLaneTrack* and the sum of the lengths of all *Transporters* located on it determine how many *Transporters* it can accommodate at any one time.

Each lane of the *TwoLaneTrack* may have its own length to realistically model the length of the lanes, when the *TwoLaneTrack* turns a corner. In that case the outside lane is longer than the inside lane.

You can change the length of the graphic and the anchor points of the *TwoLaneTrack* by clicking **Show Manipulators**  on the **Edit** ribbon tab or by pressing **M** on the keyboard.

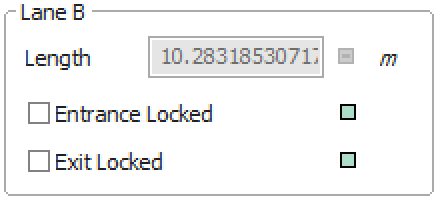
SimTalk

B.**Length [SimTalk] - lane A or B**

B.**OccupiedLength [SimTalk] - lane A or B**

Entrance Locked [lane B]

To prevent that *Transporters* enter **Lane B** of the *TwoLaneTrack*, select this check box.



Remarks

The *TwoLaneTrack* will finish transporting the *Transporters* already located on it.

It does not accept any additional *Transporters* and enters those into the **Blocking List** of **Lane B**.

- If the *Transporter* wants to enter the *TwoLaneTrack* at its entrance, it is entered into the **Forward Blocking List** of **lane B**.
- If the *Transporter* is moving backward and want to enter a length-oriented object at its exit, it is entered into the **Exit Blocking List** of **Lane B**.

Transporting starts again once you clear the check box. The first *Transporter* in the **Forward Blocking List** of **Lane B** is going to be moved on first.

SimTalk

B.**EntranceLocked [SimTalk] - lane A or B**

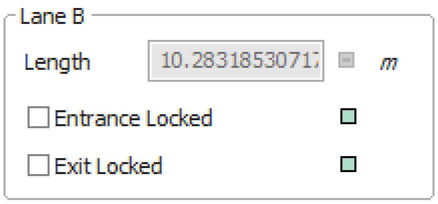
See also

Forward Blocking List

Exit Blocking List

Exit locked [lane B]

To prevent that *Transporters* exit **Lane B** of the *TwoLaneTrack*, select this check box.



Remarks

The *TwoLaneTrack* does not transport the *Transporters* to the successor in the material flow, but enters them into the **Exit Blocking List** of **Lane B** of the *TwoLaneTrack*.

Once the exit is unlocked again, the *TwoLaneTrack* moves the first *Transporter* in the **Exit Blocking List** of **Lane B** on to its successor.

> **Note**
> The *Transporter* does not trigger the **Exit Control** if you selected the check box **Exit locked**.

SimTalk

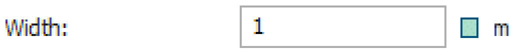
B.**ExitLocked [SimTalk] - lane A or B**

See also

Exit Blocking List

Width [text box] - TwoLaneTrack

Type the **Width** of both lanes of the *TwoLaneTrack* into the text box.

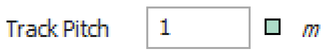


SimTalk

Width [SimTalk] - TwoLaneTrack

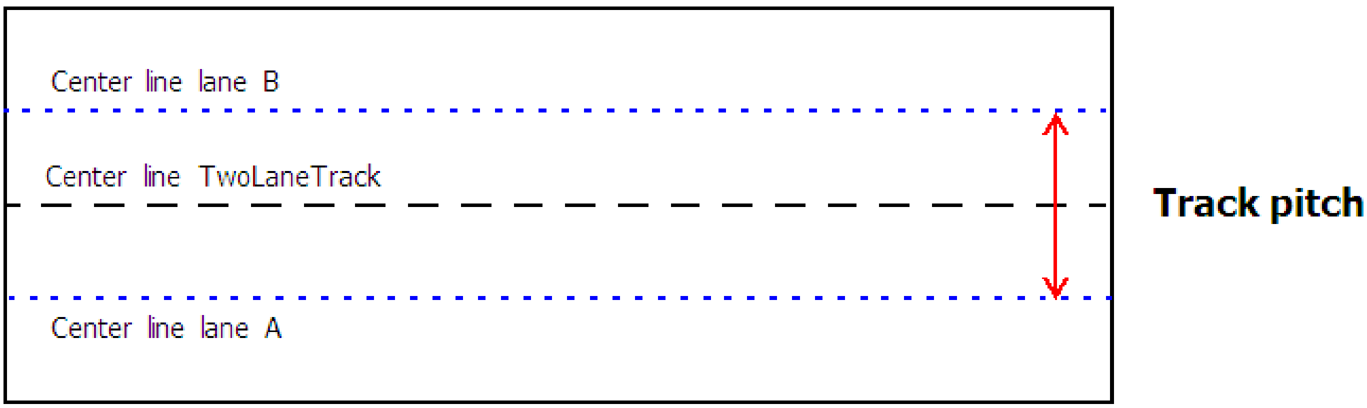
Track Pitch [text box]

Type in the **Track Pitch** between **Lane A** and **Lane B**.

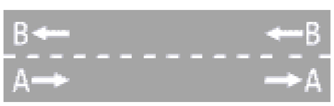
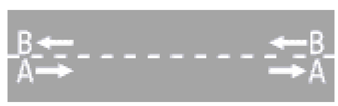


Remarks

The **Track Pitch** is the distance between the center lines of the two lanes of the two-lane track.



Compare this example:

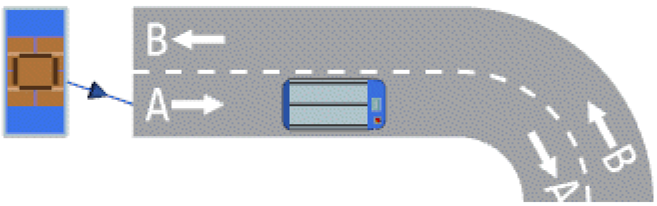
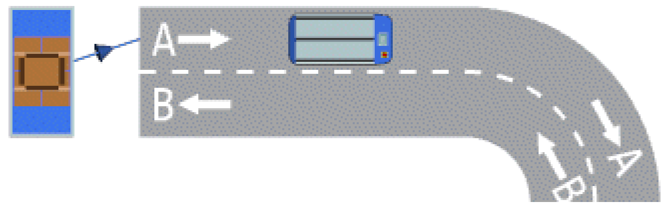
Track pitch 1	Track pitch 0.6
	

SimTalk

TrackPitch [SimTalk]

Right-hand Traffic/Left-hand Traffic

Select the side of the *TwoLaneTrack* on which traffic moves: **Right-hand Traffic** or **Left-hand Traffic**.

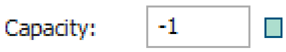
Right-hand traffic	Left-hand traffic
	

SimTalk

Traffic [SimTalk] - TwoLaneTrack

Capacity [text box] - TwoLaneTrack

Type the **Capacity** of the *TwoLaneTrack* into the text box.



Remarks

The **Capacity** is the maximum number of *Transporters* that may be located on both lanes of the *TwoLaneTrack* as a whole or in part at any one time.

The default value **-1** stands for an infinite capacity.

SimTalk

Capacity [SimTalk] - TwoLaneTrack

Destination List A [text box]

Click the ellipsis button and select a *List* object in the dialog **Select Object**.

Remarks

Type in all destination objects into this *List* object, which the *TwoLaneTrack* can reach, when the *Transporter* drives forward on **Lane A**. The forward destination list of **Lane A** concurrently is the backward destination list of **Lane B**.

Plant Simulation evaluates the destination lists when a *Transporter* is to be moved on to a successor. *Plant Simulation* searches for the destination of the *Transporter* in the destination lists of all succeeding *TwoLaneTracks*, which are directly connected with *Connectors*. If a succeeding *TwoLaneTrack* is connected via an *Interface*, this *Interface* should only have a single successor because *Plant Simulation* only searches for a destination list in the first successor of the *Interface*

SimTalk

DestListA [SimTalk] - TwoLaneTrack

See also

Select Object [for controls]

Destination List B [text box]

Click the ellipsis button and select a *List* object in the dialog **Select Object**.

Remarks

Type in all destination objects into this *List* object, which the *TwoLaneTrack* can reach, when the *Transporter* drives forward on **Lane B**. The forward destination list of **Lane B** concurrently is the backward destination list of **Lane A**.

Plant Simulation evaluates the destination lists when a *Transporter* is to be moved on to a successor. *Plant Simulation* searches for the destination of the *Transporter* in the destination lists of all succeeding *TwoLaneTracks*, which are directly connected with *Connectors*. If a succeeding *TwoLaneTrack* is connected via an *Interface*, this *Interface* should only have a single successor because *Plant Simulation* only searches in the for a destination list in first successor of the *Interface*.

SimTalk

DestListB [SimTalk] - TwoLaneTrack

See also

Select Object [for controls]

Tab Failures

Define **failures** as described under the **Tab Failures**.

Tab Controls A/B

Provides controls to modify the built-in behavior of the object.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

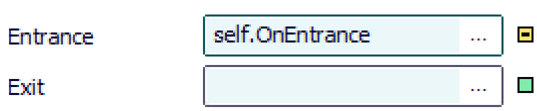
Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



  Type the source code of this control into the *Method* that opens.

  To edit the source code later on:

  - Press **F2**.
  - Or hold down **Shift** and double-click into the text box.
  - Or select **Open Object** on the context menu.
  - Or click the tab **User-defined** and double-click the name of the *Method* in the list.

- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

You can create an **Entrance Control**, an **Exit Control**, a **Backward Entrance Control**, a **Backward Exit Control**, and a **Pull Control** for **Lane A** and/or for **Lane B** of the *TwoLaneTrack*.

In addition, you can:

- Select the **Shift Calendar**.
- Create and insert **Sensors** on each lane separately or on both lanes. The *TwoLaneTrack* provides settings to accommodate the two lanes. The button **Sensors** shows how many sensors you inserted for this lane. Clicking the button shows the **Sensor List**, which displays all sensors you defined for the two lanes.

See also

Define Controls for Length-Oriented Objects

Select Object [for controls]

Entrance Control [general description]

Exit Control [general description]

Backward Entrance Control

Backward Exit Control

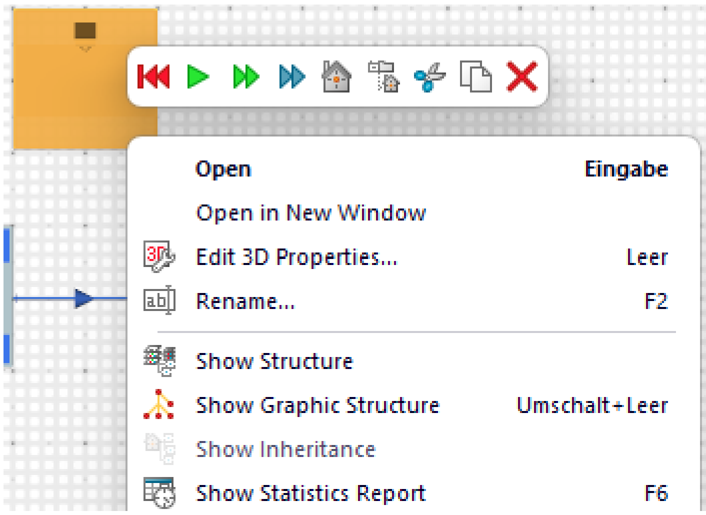
Pull Control [general description]

Shift Calendar [tab Controls]

Tab Statistics

Statistics is described under the **Tab Statistics**.

To view **Resource Statistics of Stationary Resources** in the **Statistics Report**, select **View > Show Statistics Report** in the dialog of the object. You can also click the right mouse button in the *Frame* and select **Show Statistics Report** or you can press **F6**.



See also

Resource Statistics [check box]

Resource Type

Tab User-defined

Define your own attributes as described under the **Tab User-defined**.

Navigate Menu

The commands are described under the **Navigate Menu**.

View Menu

The **View Menu** of the *TwoLaneTrack* provides commands to access the functions for each of its two lanes.

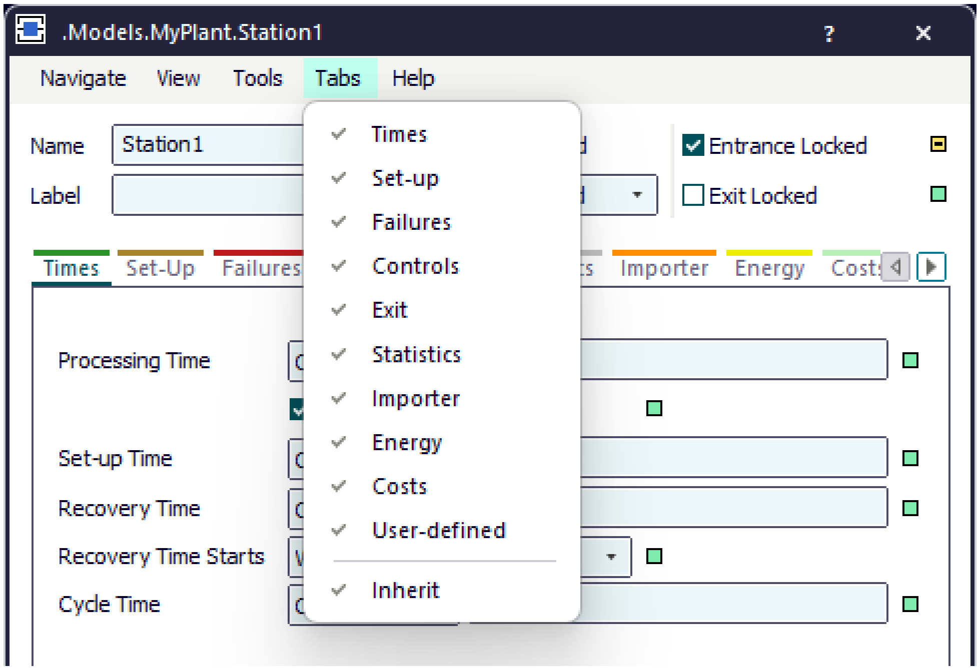
Refresh [on View menu]	Backward Blocking List (bwBlockList [SimTalk] - lane A or B)
Show Statistics Report [on View menu]	Exit Blocking List (exitBlockList [SimTalk] - lane A or B)
Show Attributes and Methods [on View menu]	Associated Lockout Zones
Contents [material flow objects] (contentsList [SimTalk] - lane A or B)	Associated Shift Calendar
Forward Blocking List (fwBlockList [SimTalk] - lane A or B)	

Tools Menu

The commands are described under the **Tools Menu**.

Tabs Menu

Use the commands of the **Tabs** menu to show or hide individual tabs of the selected material flow objects. If you hide tabs that you do not need, *Plant Simulation* opens the dialog faster, and you can change to those tabs faster that you need in your daily work.



Remarks

To apply the changed settings, click **OK**, close the dialog, and reopen it

The menu shows a check mark to the left of the displayed tabs.

The command **Inherit** turns inheritance of the displayed or hidden tabs in the dialog off or on.

Help Menu

The commands are described under the **Help Menu**.

Methods of the TwoLaneTrack

_Methods of the TwoLaneTrack

The *TwoLaneTrack* provides:

- The *methods* listed in the table of contents to the left.
- The **_Methods of Curved Objects**.
- The **Methods of the Material Flow Objects**.
- The **Methods of All Objects**.

> **Note**
> A number of methods, which the *TwoLaneTrack* shares with the other material flow objects, apply to a lane, instead of to the entire object.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.